Beyond Fixation: A Systematic Review of Peripheral Vision Use in Driving

Victor Meulenkamp[1*], Katja Kircher[2], & Christian Vater[1]

[1] Institute of Sport Science, University of Bern, Bremgartenstrasse 145, Bern, Switzerland
[2] The Swedish National Road and Transport Research Institute (VTI), S-581 95 Linkoping, Sweden
[*]victor.meulenkamp@unibe.ch

Abstract: Driving is a dynamic and sometimes complex task in which safety depends on the ability to perceive important information under time pressure. Given these constraints, peripheral vision plays a critical role in driving as it is not feasible to direct the fovea and fixate all relevant information simultaneously. This systematic review examines how perceptual, cognitive, and motor factors influence the use of peripheral vision in driving. Following PRISMA guidelines, a systematic search was conducted across PubMed, Scopus, and Web of Science. A total of 6,890 records were identified. After duplicate removal and abstract screening, 757 articles were eligible for full-text review. Studies were included if they investigated peripheral vision in dynamic driving settings and reported objective gaze measures. Eligible studies were categorized according to study design, manipulated factors, and driving task. This review will provide a structured framework for understanding peripheral vision in driving and provide a link to sport science, where athletes have to make decisions in similarly complex scenarios, also under time pressure.

1. Introduction

A driver approaches a busy intersection while maintaining lane position, monitoring a cyclist approaching from the right, checking the rear-view for overtaking vehicles, and anticipating a potential change of the traffic light, without continuously fixating each of these elements. This everyday situation illustrates the research topic of the current systematic review: the role of peripheral vision in driving. Driving is a dynamic activity in which drivers' actions not only depend on what they fixate, but also on what they perceive and monitor outside the fovea. Understanding how peripheral vision is used in driving is therefore critical for advancing research on driver distraction and inattention.

Peripheral vision refers to the visual field beyond 4 to 5 degrees around the fovea and differs from foveal vision in anatomical representation and perceptual capabilities (Larson & Loschky, 2009). Although the fovea comprises only about 1% of the retinal surface, it occupies 50% of the visual cortex (Curcio et al., 1990; Tootell et al., 1982), resulting in higher spatial resolution and color sensitivity compared to the periphery (Anstis, 1974; Strasburger et al., 2011; Abramov & Gordon, 1977; Gordon & Abramov, 1977; Hansen et al., 2009). With increasing eccentricity, visual acuity declines and object recognition becomes constrained by crowding, the phenomenon whereby objects become difficult to identify when surrounded by nearby stimuli (Bouma, 1970; Rosenholtz, 2016, 2020). These limitations can in principle be overcome by shifting gaze. However, eye movements (saccades) themselves have temporal costs and visual suppression, requiring to continuously trade off between relying on peripheral input and initiating a saccade to obtain high-resolution foveal information.

Peripheral vision is closely intertwined with attentional mechanisms, which determine how efficiently information can be extracted outside the fovea. Covert attention (i.e., attending to an object away from the point of gaze) enhances contrast sensitivity (Cameron et al., 2002; Carrasco, 2011), processing speed (Carrasco et al., 2006) and change detection in peripheral locations (Vater, 2019). The concept of the useful field of view (UFOV) captures the region around fixation from which information can be extracted without eye movements (Ringer et al., 2016). Importantly, UFOV is not fixed but can vary due to cognitive load (Gaspar et al., 2016), age (Underwood et al., 2005) and emotional state (Zhang et al., 2016). Under high attentional demands, peripheral processing efficiency can decrease, potentially narrowing the effective visual field.

Basic vision science shows that although peripheral vision is limited, it still provides useful information for many tasks. For example, it supports saccade planning, allows tracking of multiple objects at once, perceive the gist of a scene and perform object-recognition tasks(Wolfe et al., 2017). These findings suggest that peripheral vision is important for many everyday actions. However, its role is often underestimated in laboratory experiments, which typically use simplified stimuli and single-task settings that do not reflect real-world complexity. To fully understand how peripheral vision supports behavior, it is therefore necessary to examine its role in more complex and dynamic environments.

Applied research provides this broader perspective by examining peripheral vision in real-world tasks. According to the systematic review by Vater et al. (2022), peripheral vision supports four key functions across daily life tasks such as driving. These functionalities are monitoring the environment, presaccadic preview, managing saccade and eccentricity costs, and supporting action planning. Monitoring refers to the continuous extraction of information from the surroundings without direct fixation, such as detecting lane boundaries. Presaccadic preview allows observers to gather preliminary information about a peripheral object before fixating it, thereby facilitating efficient saccades. Finally, peripheral vision provides information that supports action planning, such as maintaining lane position through optic flow cues.

Although perceptual, cognitive and, motor constraints are well established in basic science, their combined

influence on peripheral vision use in dynamic driving remains unclear. Moreover, many driving studies rely on eye movement measures and assume that fixation reflects information processing (Harbluk et al., 2007; Strayer & Johnston, 2001), potentially underestimating the contribution of peripheral vision (Rosenholtz, 2016). Additionaly, unlike in sports research, where functional gaze strategies (e.g., gaze anchor, visual pivot, foveal spot) have been systematically described, no comparable framework exists to characterize patterns of peripheral vision usage in traffic contexts (Vater et al., 2020).

The present systematic review addresses these gaps by identifying and categorizing factors that influence peripheral vision use in driving. Specifically, we examine how perceptual, cognitive, and motor factors influence peripheral vision use in driving and how this affects gaze behavior and driver performance. By integrating findings across experimental paradigms and driving contexts, this review aims to provide a systematic framework that can be used in future research on driver distraction and inattention.

2. Method

2.1 Identification

We followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) procedure (Moher et al., 2009). The database search was conducted in April 2025 using Pubmed, Scopus and Web of Science. Only peer-reviewed articles written in English with accessible full texts were considered. Search terms were defined a priori and applied to titles, abstracts, and where available, keywords. A detailed description of the search strategy is provided in Appendix A (Table 1).

2.2 Screening

Using the predefined search strategy, a total of 6,890 records were identified from the selected databases. After removal of 2,089 duplicates, 4,801 records remained for title and abstract screening. The screening was conducted following a structured sequence based on the predefined criteria (see Appendix A, Table 1). First, records were screened for relevance to the target population, excluding studies involving non-human subjects (e.g., animals) and non-driver populations (e.g., cyclists, pedestrians, motorcyclists). Second, methodological criteria were applied, excluding studies conducted in inappropriate settings. Only studies involving dynamic driving-related visual content (i.e., real world driving, driving simulator, or screen-based/video-based driving contexts) were included, whereas static image tasks were excluded. Third, studies were assessed for the presence of objective gaze measures. Studies that did not include any form of gaze assessment were excluded. This process resulted in 757 records eligible for full-text screening.

2.3 Eligibility

Full texts were retrieved and screened according to more specific eligibility criteria focusing on the role of peripheral vision in driving. Studies were excluded if they focused solely on foveal vision without reference to peripheral vision. To ensure the relevance, the discussion section of each paper was screened for a list of terms, see Appendix A (Table 1). Studies that didn't address one of these terms were excluded. In addition, studies had to fit within the predefined categorization framework (study type, manipulation type, and driving task; see Appendix A, Table 2 & 3). Articles that could not be categorized across all three levels were excluded. So far, we excluded 526 full texts based on these criteria and will finish the screening by the end of May 2026. We expect to include 150 to 200 papers in this systematic review.

3. Preliminary results

The first results of the full-text screening indicate that driving requires close coordination of perceptual, cognitive, and motor processes. Drivers maintain lane position using peripheral optic flow and road-edge information (Kountouriotis et al., 2011), detect hazards that initially appear outside fixation (Shinoda et al., 2001), monitor mirrors while keeping gaze forward (Doshi & Trivedi, 2012), and respond to sudden events before fixating them (Huestegge & Böckler, 2016). Importantly, applied findings indicate that drivers sometimes initiate braking or steering responses prior to directly fixating a hazard, demonstrating that peripheral information can trigger action. Moreover, expertise and cognitive load influence how drivers balance peripheral monitoring and eye movements. Experienced drivers differ in their distribution of attention and gaze behavior compared to novices (Lamble et al., 1999; Summala et al., 1996; Underwood et al., 2003), and higher cognitive workload has been associated with reduced peripheral monitoring efficiency and functional visual narrowing, often described as visual tunnelling (Gaspar et al., 2016; Victor et al., 2005). These findings explain that peripheral vision plays a central role in safe driving performance. The full analysis of the included articles will provide a qualitative and quantitative summary of identified functionalities and discuss potential mechanisms linked to perceptual, cognitive and motor factors.

References

Abramov, I., & Gordon, J. (1977). Color vision in the peripheral retina. I. Spectral sensitivity. *Journal of the Optical Society of America*, *67*(2), 195–202. https://doi.org/10.1364/josa.67.000195

Anstis, S. M. (1974). A chart demonstrating variations in acuity with retinal position. *Vision Research*, *14*(7), 589–592. https://doi.org/10.1016/0042-6989(74)90049-2

Bouma, H. (1970). Interaction effects in parafoveal letter recognition. *Nature*, *226*(5241), 177–178. https://doi.org/10.1038/226177a0

Cameron, E. L., Tai, J. C., & Carrasco, M. (2002). Covert attention affects the psychometric function of contrast sensitivity. *Vision Research*, *42*(8), 949–967. https://doi.org/10.1016/s0042-6989(02)00039-1

Carrasco, M. (2011). Visual attention: The past 25 years. *Vision Research*, *51*(13), 1484–1525. https://doi.org/10.1016/j.visres.2011.04.012

Carrasco, M., Giordano, A. M., & McElree, B. (2006). Attention speeds processing across eccentricity: Feature and conjunction searches. *Vision Research*, *46*(13), 2028–2040. https://doi.org/10.1016/j.visres.2005.12.015

Curcio, C. A., Sloan, K. R., Kalina, R. E., & Hendrickson, A. E. (1990). Human photoreceptor topography. *The*

Journal of Comparative Neurology, *292*(4), 497–523. https://doi.org/10.1002/cne.902920402

Doshi, A., & Trivedi, M. M. (2012). Head and eye gaze dynamics during visual attention shifts in complex environments. *Journal of Vision*, *12*(2), 9. https://doi.org/10.1167/12.2.9

Gaspar, J. G., Ward, N., Neider, M. B., Crowell, J., Carbonari, R., Kaczmarski, H., Ringer, R. V, Johnson, A. P., Kramer, A. F., & Loschky, L. C. (2016). Measuring the Useful Field of View during Simulated Driving with Gaze-Contingent Displays. *Human Factors*, *58*(4), 630–641. https://doi.org/10.1177/0018720816642092

Gordon, J., & Abramov, I. (1977). Color vision in the peripheral retina. Ii. Hue and saturation. *Journal of the Optical Society of America*, *67*(2), 202–207. https://doi.org/10.1364/josa.67.000202

Hansen, T., Pracejus, L., & Gegenfurtner, K. R. (2009). Color perception in the intermediate periphery of the visual field. *Journal of Vision*, *9*(4), 26. https://doi.org/10.1167/9.4.26

Harbluk, J. L., Noy, Y. I., Trbovich, P. L., & Eizenman, M. (2007). An on-road assessment of cognitive distraction: Impacts on drivers' visual behavior and braking performance. *Accident Analysis and Prevention*, *39*(2), 372–379. https://doi.org/10.1016/j.aap.2006.08.013

Huestegge, L., & Böckler, A. (2016). Out of the corner of the driver's eye: Peripheral processing of hazards in static traffic scenes. *Journal of Vision*, *16*(2), 11. https://doi.org/10.1167/16.2.11

Kountouriotis, G. K., Floyd, R. C., Gardner, P. H., Merat, N., & Wilkie, R. M. (2011). The role of gaze and road edge information during high-speed locomotion. *Journal of Experimental Psychology. Human Perception and Performance*, *38*(3), 687–702. https://doi.org/10.1037/a0026123

Lamble, D., Laakso, M., & Summala, H. (1999). Detection thresholds in car following situations and peripheral vision: Implications for positioning of visually demanding in-car displays. *Ergonomics*, *42*(6), 807–815. https://doi.org/10.1080/001401399185306

Larson, A. M., & Loschky, L. C. (2009). The contributions of central versus peripheral vision to scene gist recognition. *Journal of Vision*, *9*(10), 6. https://doi.org/10.1167/9.10.6

Moher, D., Liberati, A., Tetzlaff, J., Altman, D. G., Antes, G., Atkins, D., Barbour, V., Barrowman, N., Berlin, J. A., Clark, J., Clarke, M., Cook, D., D'Amico, R., Deeks, J. J., Devereaux, P. J., Dickersin, K., Egger, M., Ernst, E., Gøtzsche, P. C., … Tugwell, P. (2009). Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *PLoS Medicine*, *6*(7), e1000097. https://doi.org/10.1371/journal.pmed.1000097

Ringer, R. V., Throneburg, Z., Johnson, A. P., Kramer, A. F., & Loschky, L. C. (2016). Impairing the useful field of view in natural scenes: Tunnel vision versus general interference. *Journal of Vision*, *16*(2), 7. https://doi.org/10.1167/16.2.7

Rosenholtz, R. (2016). Capabilities and Limitations of Peripheral Vision. *Annual Review of Vision Science*, *2*, 437–457. https://doi.org/10.1146/annurev-vision-082114-035733

Rosenholtz, R. (2020). Demystifying visual awareness: Peripheral encoding plus limited decision complexity resolve the paradox of rich visual experience and curious perceptual failures. *Attention, Perception & Psychophysics*, *82*(3), 901–925. https://doi.org/10.3758/S13414-019-01968-1

Shinoda, H., Hayhoe, M. M., & Shrivastava, A. (2001). What controls attention in natural environments? *Vision Research*, *41*(25–26), 3535–3545. https://doi.org/10.1016/S0042-6989(01)00199-7

Strasburger, H., Rentschler, I., & Jüttner, M. (2011). Peripheral vision and pattern recognition: A review. *Journal of Vision, 11*(5), 13. https://doi.org/10.1167/11.5.13

Strayer, D. L., & Johnston, W. A. (2001). Driven to distraction: Dual-task studies of simulated driving and conversing on a cellular telephone. *Psychological Science*, *12*(6), 462–466. https://doi.org/10.1111/1467-9280.00386

Summala, H., Nieminen, T., & Punto, M. (1996). Maintaining lane position with peripheral vision during in-vehicle tasks. *Human Factors*, *38*(3), 442–451. https://doi.org/10.1518/001872096778701944

Tootell, R. B. H., Silverman, M. S., Switkes, E., & De Valois, R. L. (1982). Deoxyglucose analysis of retinotopic organization in primate striate cortex. *Science (New York, N.Y.)*, *218*(4575), 900–901. https://doi.org/10.1126/SCIENCE.7134981

Underwood, G., Chapman, P., Brocklehurst, N., Underwood, J., & Crundall, D. (2003). Visual attention while driving: Sequences of eye fixations made by experienced and novice drivers. *Ergonomics*, *46*(6), 629–646. https://doi.org/10.1080/0014013031000090116

Underwood, G., Phelps, N., Wright, C., Van Loon, E., & Galpin, A. (2005). Eye fixation scanpaths of younger and older drivers in a hazard perception task. *Ophthalmic and Physiological Optics*, *25*(4), 346–356. https://doi.org/10.1111/j.1475-1313.2005.00290.x

Vater, C. (2019). How selective attention affects the detection of motion changes with peripheral vision in MOT. *Heliyon*, *5*(8), e02282. https://doi.org/10.1016/J.HELIYON.2019.E02282

Vater, C., Williams, A. M., & Hossner, E. J. (2020). What do we see out of the corner of our eye? The role of visual pivots and gaze anchors in sport. *International Review of Sport and Exercise Psychology*, *13*(1), 81–103. https://doi.org/10.1080/1750984X.2019.1582082

Vater, C., Wolfe, B., & Rosenholtz, R. (2022). Peripheral vision in real-world tasks: A systematic review. *Psychonomic Bulletin & Review*, *29*(5), 1531–1557. https://doi.org/10.3758/s13423-022-02117-w

Victor, T. W., Harbluk, J. L., & Engström, J. A. (2005). Sensitivity of eye-movement measures to in-vehicle

task difficulty. *Transportation Research Part F: Traffic Psychology and Behavior*, *8*(2), 167–190. https://doi.org/10.1016/j.trf.2005.04.014

Wolfe, B., Dobres, J., Rosenholtz, R., & Reimer, B. (2017). More than the Useful Field: Considering peripheral vision in driving. *Applied Ergonomics*, *65*, 316–325. https://doi.org/10.1016/j.apergo.2017.07.009

Zhang, T., Chan, A. H. S., Ba, Y., & Zhang, W. (2016). Situational driving anger, driving performance and allocation of visual attention. *Transportation Research Part F: Traffic Psychology and Behaviour*, *42*(2), 376–388. https://doi.org/10.1016/j.trf.2015.05.008